Valuing Mangrove Loss: Evidence from a Difference-in-Differences Study of Hurricane Irma and Housing Prices in Collier County

I. Introduction

Mangroves are increasingly recognized as a critical component of coastal protection. Studies demonstrate that mangroves and coastal wetlands provide flood-risk reduction worldwide, preventing over $65 billion in flood damages and reducing risk for more than 15 million people each year (Menéndez et al., 2020). In addition, mangroves contribute to climate mitigation through carbon sequestration and support local economies through ecotourism (Choudhary et al., 2024; Blanton et al., 2024). However, these benefits are not traded in markets and remain undervalued in ecosystem services.

Evidence consistently shows the critical protective functions that mangroves provide. In Collier County, southwestern Florida, mangroves reduce property surge losses by $67 million annually (Narayan et al., 2025) and numerical models demonstrate their effectiveness at reducing surge heights during fast-moving storms (around 40 km/hr) (Narayan et al., 2017). In the Sundarban region of Bangladesh, the protection value of mangroves is estimated at $543.30 million using the Damage-Cost-Avoided (DCA) method (Sarkar et al., 2020). Yet the failure to properly value these protective ecosystem services has contributed to ongoing mangrove loss and under-investment in restoration. Mangrove areas have declined from approximately 139,777 km² in 2000 to 131,931 km² in 2014 (Hamilton and Casey, 2016). In response, initiatives such as the UN System of Environmental-Economic Accounting (SEEA) and the World Bank's natural capital programs have called for integrating ecosystem service values into national statistics and cost–benefit analyses. Several studies also highlight how mangroves help to shield communities and reduce damages during storms and tsunamis (Das & Vincent, 2009; Marois & Mitsch, 2015; Kelty et al., 2022). However, many existing valuations do not measure how people's revealed willingness-to-pay for protection is reflected in real markets.

Theoretically, mangrove protection that provides economic benefits to homeowners should be capitalized into housing prices. Houses in high-risk flood zones typically sell at price discounts relative to comparable houses outside floodplains (Bin et al., 2008). Homebuyers are also forward-looking and rational, responding more strongly to risks they have directly observed or visualized (Atreya & Ferreira, 2015). Evidence from the removal of insurance premium subsidies shows that buyers capitalize disaster-related costs into property prices (Georgic & Klaiber, 2022). Based on this, we hypothesize that the protective benefits of mangrove forests are fully capitalized into housing prices only after a storm creates an information shock that demonstrates their value.

Hurricane Irma offers such a case. The storm brought coastal flooding and building damages in Collier County with peak wind speeds reaching Category 3 up to 112 mph or 50 m/s (Paramygin & Sheng, 2025). Collier County is also bordered by some of the largest continuous mangrove tracts in the continental United States. Lagomasino et al. (2021) detects approximately 10,760 ha of mangroves showed evidence of complete dieback within 15 months after the storm, primarily due to storm-surge flooding and saltwater ponding. In contrast, mangroves in well-drained areas recovered more quickly.

This difference in mangrove survival across space provides a quasi-experimental setup to compare similar neighborhoods that were exposed to the hurricane. Properties near mangroves that died off form our treatment group, while those protected by surviving mangroves serve as the control group. We employ a hedonic pricing model in a difference-in-differences (DiD) design to compare property value changes between these groups before and after Irma. By including location fixed effects to absorb time-invariant spatial unobservables (Linden & Rockoff, 2008; Muehlenbachs et al., 2015) and adapting the hedonic DiD framework utilized by Shr et al. (2023), we control for baseline price levels and market trends. Through this framework, we aim to identify how much of mangroves' storm-protection value is reflected in coastal property prices once a hurricane reveals the magnitude of their risk-reduction services.

II. Literature Review

Mangrove forests are recognized as natural defences in coastal areas. Research by Costanza et al. (2008) and Menéndez et al.(2020) confirms that mangrove ecosystems provide billions of dollars in annual storm protection. In the specific context of Florida, mangroves averted \$1.5 billion in storm damages during Hurricane Irma alone, amounting to a 25% savings in counties with mangrove coverage and helping protect more than 626,000 people across Florida (Narayan et al., 2019).

In housing markets, these protective services are not directly priced but are instead capitalized into property values through characteristics that buyers implicitly pay for. It has been observed that consumers make decisions not by a single character but by the number of characteristics (Xiao, 2017). Recent literature argues that this capitalization is dynamic rather than static. Atreya and Ferreira (2015) and Cohen et al. (2021) provide compelling evidence of an information or surprise effect shows that markets often undervalue latent risks until a disaster makes them salient. Any market premium for natural defenses like mangroves would likely become visible only after a significant environmental shock when buyers reassess the necessity of protection. Supporting this idea, Hino & Buker (2021) using data on millions of home sales find that lack of information appears to contribute to underpricing real estate.

Despite these advances, the specific question of how mangrove protection is incorporated into property values remains insufficiently studied. We identified only one working paper that uses a hedonic pricing model to examine how mangroves affect housing prices in Florida (Liu et al., 2025). Their study estimates the willingness-to-pay (WTP) for mangrove protection by analyzing repeat home sales and comparing prices based on proximity to mangroves. However, they do not use a difference-in-differences (DiD) framework to compare outcomes before and after a major hurricane. Without that, their estimates may suffer from omitted variable bias and cannot isolate causal effects. Moreover, Himes-Cornell et al.

(2018) showing that between 2007 and 2016, relatively few mangrove ecosystem service valuation studies used revealed preference methods (Fig. 1).

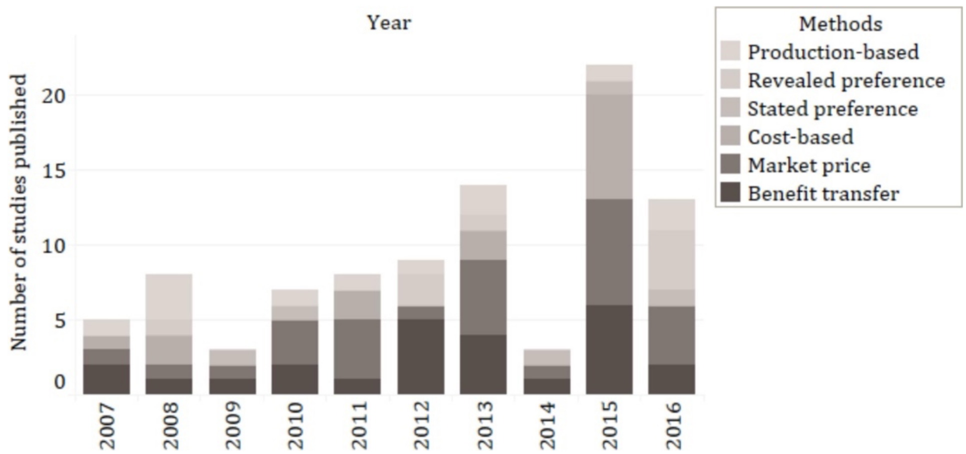

Fig. 1. Trends in mangrove valuation methods 2007–2016. Adapted from Cornell et al. (2018).

This study seeks to address these gaps. We adopt a revealed preference approach within the Americas and leverage the shock of Hurricane Irma in a difference-in-differences design. The approach follows recent work in environmental economics by Shr et al. (2023) which used a similar DiD strategy. Through this approach, our research will causally identify whether housing markets actually recognize and price the storm-protection services provided by mangrove forests.

III. Conceptual and Theoretical Framework

1. Hedonic Valuation of Mangroves as Natural Capital and Risk-Reducing Infrastructure

The hedonic pricing method (HPM) derived from Lancaster (1966) consumer theory and Rosen (1974) model has been used extensively to analyze housing markets. Under this framework, a home is viewed as a bundle of characteristics that contribute to its market price where buyers reveal their willingness to pay for these attributes through observed transactions. A broad literature demonstrates that housing prices respond systematically to environmental hazards and risk exposure (Fuerst & Myers, 2021; Nicholls, 2019; Dai et al., 2020). For instance, Bin and Landry (2013) found that after two major hurricanes in North Carolina, floodplain properties sold 5.7–8.8% lower than comparable homes outside flood zones that indicate disasters act as information shocks that correct homeowners' prior underestimation of hazard risk.

Within this hedonic framework, coastal mangrove forests can be understood as both an amenity and a risk-reducing asset for nearby properties. A home located behind an intact mangrove belt enjoys an implicit insurance benefit in the form of lower expected storm-surge damage. However, mangrove protection behaves as a form of natural capital whose value depreciates when the ecosystem is damaged or destroyed. A hurricane landfall acts as a stress test or a sudden depreciation shock to this natural asset. Property behind an intact mangrove belt retains a different risk profile than identical property where the mangrove barrier has collapsed. This difference fundamentally alters the long-run environmental protection embedded in the property's location.

In principle, this protective benefit should be capitalized into property values in the same way that housing markets capitalize other nature-based solutions (NbS). Empirical studies support the view that nature-based solutions (NbS) are capitalized into real estate values. The implementation of a major NbS project in the Netherlands showed that such amenities provide roughly a 15% premium (approximately €33,687) to nearby residential property prices (Mutlu et al., 2023). Specific to the context of Florida, Liu et al. (2025) analyzed housing transactions and found that proximity to mangroves significantly helped reduce property value losses following hurricanes. Homes located near mangroves were 3 to 4 percentage points less likely to experience major price declines. Property valued at $1 million translates to an estimated benefit of approximately $20,000 to $40,000.

Building on these insights, our study focuses on the specific stress test provided by Hurricane Irma in 2017 in Collier County, Florida. Before a major storm, homeowners may take the presence of nearby mangroves for granted, meaning market prices might not fully reflect their insurance benefit. However, by exploiting the spatial variation in mangrove damage caused by Irma where some stands survived while others failed, we aim to isolate the storm risk–reduction component of mangrove value. This approach allows us to observe how the market re-prices properties once the reliability of their natural defense is revealed.

2. Behavioral Mechanism

While hedonic theory provides the economic structure for how environmental risk should be priced into assets, behavioral and informational mechanisms explain why and when these risks become reflected in market prices. A key insight from behavioral economics is that extreme events can cause people to update their beliefs about risk, especially when those events are highly salient. Stimulus-driven theories argued that, irrespective of the goals and task of the observer, salient objects capture attention (Theeuwes, 1992, 2010). A landfalling hurricane is a prime example of a salient event that captures the attention of homeowners to confront risks that they may have previously underestimated or even ignored.

In the context of disasters and risk perception, two key belief-updating mechanisms are frequently discussed. The first is the availability heuristic, a cognitive shortcut through which individuals estimate the probability of an event based on how easily similar examples come to mind (Tversky & Kahneman, 1973). A severe hurricane that caused extensive damage will be fresh in memory which makes people temporarily perceive a higher risk of similar events. Under this heuristic-driven response, changes in risk perception may be large but short-lived as the collective memory of the disaster gradually fades and normal conditions eventually return. Many studies also investigate the persistence of flood risk discounts over time. Bin and Landry (2013) report that the price discount eventually disappeared five or six years after the flood in Pitt County, North Carolina, the US. Similarly, Atreya et al. (2013) find that the negative effect of the flood decays rapidly, vanishing four to nine years after the flood in Dougherty County, Georgia, US.

The second mechanism is Bayesian learning, where individuals update their risk beliefs by incorporating credible, location-specific information. In the context of a hurricane, this means observing whether nearby mangroves withstood storm surge or failed. These outcomes are visible and persistent, they act as strong signals that can shift long-term expectations about protective effectiveness. Unlike short-term emotional responses, Bayesian updates are durable and rational. This aligns with Kousky (2010) who found that individuals adjust flood risk perceptions when exposed to clear, geographically relevant disaster information.

In our study, the Bayesian learning mechanism is likely the dominant driver of market behavior. The post-Irma landscape in Collier County provided a continuous and visible signal of mangrove infrastructure condition. This signal is literally embedded in the local geography, homebuyers touring an area can see whether the mangroves are healthy or standing dead. Unlike a short-lived emotional reaction, this type of information does not fade quickly and instead remains available to anyone evaluating the surrounding properties. Thus, buyers are able to update their risk expectations rationally and in a location-specific manner. We theorize that as a result of this learning process, properties near destroyed mangrove forests will experience a persistent increase in perceived vulnerability, whereas properties near surviving mangroves will enjoy a reinforced perception of safety. These divergent post-storm perceptions form the theoretical basis for long-run differences in housing prices that we aim to detect.

IV. Data and Research Design

Evaluating the economic impact of mangrove protection requires combining detailed property-level transaction data with high-resolution ecological and storm-impact datasets. Our primary dataset is a longitudinal panel of residential property sales in Collier County from 2012 to 2022, obtained from the Collier County Property Appraiser and supplemented with commercial datasets such as CoreLogic or Zillow's ZTRAX. Each record includes the sale price, sale date, interior square footage, lot size, building age, number of bedrooms and bathrooms, and major amenities. Geocoded parcel coordinates allow us to spatially link each transaction to its surrounding environmental features. Thus, the unit of analysis is an individual property sale at time t that can be matched to both pre- and post-hurricane ecological conditions.

To classify mangrove conditions, we integrate data from Global Mangrove Watch with NASA/USGS remote sensing products including Landsat imagery, LiDAR-based elevation models, and hyperspectral surveys. These ecological datasets allow us to identify where mangrove stands experienced complete dieback after Hurricane Irma and where they survived. We restrict the analytic sample to coastal homes that plausibly benefited from mangrove protection prior to the hurricane, excluding inland properties outside the effective influence zone. Each home within this zone is then assigned a treatment indicator $DeadMangrove_i = 1$ if the mangrove stand died, and 0 otherwise. Because both groups possessed mangrove protection before Irma, this classification isolates homes that experienced an loss of natural protection from those that retained it.

Our analysis focus to isolates the effect of mangrove loss rather than direct hurricane damage, we try incorporate storm-impact controls from NOAA, the National Weather Service, and FEMA, including peak storm-surge depth, local flood extent, neighborhood-level wind speeds, and FEMA flood-zone designations (VE, AE, X-shaded). Including these controls accounts for baseline risk exposure and variation in physical storm damage unrelated to mangrove dieback. Distance to the coastline controls is added for baseline spatial variation in flood and storm surge exposure that existed before Irma. This ensures that our estimated effects are not confounded by naturally higher risk levels for properties located closer to the shoreline.

Hurricane Irma creates exogenous ecological variation that supports causal identification. While the hurricane broadly impacted Collier County, the spatial pattern of mangrove mortality was driven by storm-surge ponding, geomorphology, and micro-topography rather than by economic attributes of nearby properties. As a result, properties in the range of destroyed mangroves experienced a sudden depreciation of their natural capital, while otherwise similar nearby homes saw their protective infrastructure remain intact. This ecological heterogeneity allows Irma to function as a natural experiment for treated and control properties that were exposed to the same hurricane. Thus, any subsequent divergence in price trajectories can be interpreted as a market response to the loss of mangrove protection rather than to the hurricane itself.

We estimate this effect using a difference-in-differences (DiD) design with property-level panel data. The main regression specification is:

$$\ln(\text{Price}_{it}) = \alpha + \beta_1\, DeadMangrove_i + \beta_2\, Post_t + \beta_3\, (DeadMangrove_i \times Post_t) + \gamma\, X_i + \lambda_i + \tau_t + \varepsilon_{it}$$

where $Post_t = 1$ for sales after September 2017. The interaction term captures the differential post-Irma change in prices for treated properties. The vector X_i includes home's structural attributes. The dependent variable is the natural log of inflation-adjusted sale price. Property fixed effects λ_i absorb all time-invariant characteristics of each home local environment such as elevation, construction quality, and persistent neighborhood attributes. Time fixed effects τ_t (year–quarter dummies) capture market-wide trends, seasonality, macroeconomic shocks, and region-wide post-Irma effects. This structure ensures that identification arises solely from within-property changes over time and from differential exposure to mangrove dieback.

Before estimating the DiD, we empirically determine the spatial range within which mangroves influence housing prices. Following a boundary-identification strategy, we estimate hedonic regressions for the pre- and post-Irma periods:

$$\ln(\text{Price}_{it}) = \alpha + \gamma X_i + \lambda_i + \tau_t + \varepsilon_{it}$$

We then extract the residuals $\widehat{\varepsilon_{it}}$ which represent the portion of house prices not explained by housing attributes or location fundamentals. These residuals may still capture a mix of unobserved factors including mangrove proximity. We plot these residuals against the distance bins from each property to the mangrove edge separately for the pre- and post-Irma periods. If mangroves provide storm-protection value, the pre- and post-Irma residual curves will differ for homes close to mangroves but converge as distance increases. The distance at which the two curves meet indicating that mangrove proximity no longer affects prices that defines the maximum spatial range of mangrove influence. Properties beyond this threshold show no meaningful price response to mangrove conditions and are excluded from the analytic sample. This boundary-identification approach ensures that the DiD estimator focuses exclusively on homes that plausibly benefited from mangrove protection before Irma to strengthen causal interpretation of the treatment effect.

We also estimate a baseline pre-Irma hedonic model:

$$\ln(\text{Price}_i) = \alpha + \sum_b \beta_b \; DistanceBin_{ib} + \gamma X_i + \lambda_i + \varepsilon_i$$

where $DistanceBin_{ib}$ denotes a set of indicator variables for the distance ranges between each property and the nearest mangrove edge (e.g., 0–100 m, 100–200 m, 200–300 m) with one bin omitted as the reference category. This regression uses only pre-Irma sales, so it does not include a time variable as there is no variation in treatment or mangrove conditions before the storm. The purpose of this model is to determine whether homes closer to mangroves were priced higher, lower, or similarly before Irma. Identifying the pre-storm price gradient helps us assess whether mangroves were capitalized as amenities, viewed as nuisances, or not priced at all prior to the hurricane. The baseline will be important for interpreting the post-Irma treatment effects in the DiD analysis.

To validate the DiD assumptions, we estimate an event-study specification for parallel trends:

$$\ln(\text{Price}_{it}) = \alpha + \sum_{l \neq -1} \theta_l \; \{DeadMangrove_i \times 1\,(t - T_0 = l)\} + \lambda_i + \tau_t + \varepsilon_{it}$$

where T_0 denotes the quarter of Hurricane Irma's landfall and the term $1[1\,(t - T_0 = l)$ is a dummy variable equal to 1 only when a sale occurs exactly l periods (quarters) before or after Irma, and 0 otherwise. Each value of l therefore corresponds to one distinct lead or lag period relative to the storm. We omit the dummy for l = -1 which the final pre-Irma quarter and becomes the reference period.

The interaction term $\{DeadMangrove_i \times 1\,(t - T_0 = l)\}$ identifies treated properties observed in period l. Thus, each coefficient θ_l measures the difference in log sale prices between treated and control homes in quarter l relative to the last pre-Irma quarter. This structure allows us to visually and statistically examine the pre-treatment trends. If the coefficients for $l < 0$ are close to zero and statistically insignificant, this indicates that treated and control homes followed parallel trends before Irma. The coefficients for $l > 0$ then trace out the dynamic adjustment of housing markets after mangrove loss revealing whether the price penalty emerges immediately, grows over time, or gradually dissipates as risk perceptions update or as ecosystems recover. Plotting θ_l with confidence intervals provides a transparent visualization of identification quality and the temporal evolution of the treatment effect.

We conduct several robustness checks to strengthen causal interpretation. First, we compare pre-storm characteristics across treated and control properties; if substantial imbalances appear, we apply propensity-score matching to achieve better comparability. Second, we test continuous measures of treatment intensity such as the percentage of mangrove cover lost or distance to the nearest dieback zone to identify potential dose–response patterns. Third, all spatial datasets including mangrove mortality maps, storm surge and wind-speed rasters, elevation models, and parcel boundaries are integrated in a Geographic Information System (GIS) to precisely assign environmental conditions to each home. We will also produce high-resolution maps illustrating treated and control zones and associated property locations to visualize the natural experiment.

By combining detailed property data, high-resolution ecological information, and an exogenous shock generated by Hurricane Irma, this design provides a credible framework for estimating the causal effect of mangrove loss on housing prices. The DiD estimator β_3 and the dynamic event-study coefficients θ_l allow us to quantify how the housing market reprices natural protective capital when it suddenly depreciates, thereby revealing the economic value of mangrove storm-protection services.

V. Preliminary Supposition and Implications

Before running the empirical analysis, we expect several patterns to emerge based on existing theory and evidence. The impact of losing mangrove protection is likely to vary across locations. Homes positioned directly behind mangroves that died off should experience the strongest effects because they lose their immediate natural barrier against storm surge, while properties farther inland may exhibit a weaker response. Our focus is not on tracing the spatial gradient of mangrove protection, but rather to determine whether the loss of mangroves leads to measurable price differences for homes that previously benefited from their presence.

We also anticipate that price effects will change over time. After Hurricane Irma, housing markets may reflect temporary disruptions related to physical damage and uncertainty. As the range moves into the medium run and repairs are completed, price movements are expected to reflect updated beliefs about long-term storm risk. Homes that lost their mangrove buffer may experience persistent price reductions relative to comparable homes whose mangroves survived. This price gap may widen as homeowners and buyers incorporate new information about vulnerability and later narrow if memories fade or if no additional storms occur. The event-study model allows us to trace these dynamic adjustments while verifying whether treated and control homes followed similar pre-Irma trajectories. Differences across property types like as higher-value coastal homes may show larger price responses due to their greater exposure and higher expected losses, while smaller or inland homes may respond less strongly. Insurance coverage may further moderate these outcomes; insured homeowners might feel partially protected and exhibit smaller price changes, whereas uninsured homes may show larger declines.

Based on prior studies, we expect the treatment effect β_3 to be economically meaningful. Flood-risk research typically finds 3–4% price discounts after major storms (Liu et al., 2025). If β_3 lies within this range, a home valued at \$1,000,000 before Irma could lose \$20,000–\$40,000 when nearby mangroves die. The community-level impact can be expressed using standard hedonic practice as:

$$\text{Total Capitalized Loss} = |\ \beta_3\ | \times (\text{Average home price}) \times (\text{Number of treated homes})$$

This estimated loss reflects the storm-protection value recognized by the housing market. It does not capture the full ecological value of mangroves, such as carbon storage, biodiversity, and recreation, but it provides a concrete estimate of the benefit that directly affects homeowners and local governments. A significant and negative β_3 would indicate that households recognize and value the risk-reduction benefits of mangroves and adjust their willingness to pay once the storm reveals the protective function of these ecosystems. A small or insignificant β_3 would point toward under-recognition of natural protection or reliance on alternative mitigation measures such as insurance, disaster aid, or engineered defenses. A strong negative β_3 would highlight that conserving or restoring mangroves helps maintain property values and reduce future disaster costs. Local planners could integrate this information into zoning, land-use decisions, and living shoreline projects. Insurers and lenders may also incorporate the findings into risk pricing. Even if the effect is smaller than expected, the result still useful to highlights that natural protection may be undervalued and that communities may benefit from improved risk communication and greater integration of ecosystem services into decision-making.

VI. Conclusion

This proposal outlines a strategy to measure the housing-market value of mangrove storm-protection services. We focus on Hurricane Irma's impact in Florida as a natural experiment that revealed and ask how the housing market responded to that revelation. By comparing price trajectories for homes behind mangroves that died versus those behind mangroves that survived, our difference-in-differences design isolates the causal effect of losing natural protection. This quasi-experimental approach is necessary because controlled experiments are impossible in this context and the spatially uneven ecological shock offers credible treated and control groups. The use of fixed effects and granular controls further strengthens the approach.

A key part of this study is the information effect. Even though mangroves protected homes before Irma, many homeowners may not have fully realized or appreciated that protection. A major storm changes this by making the role of mangroves suddenly visible. When homeowners observe that areas with lost mangroves experience more flooding or higher damage, they update their beliefs about long-term risk. This shift in risk perception can lead to lasting price changes. By capturing how the market reacts after the storm, our study helps explain not only the physical value of mangroves but also how people become aware of that value only after seeing what happens when mangroves fail.

As coastal hazards intensify under climate change, understanding the value of nature-based defenses is increasingly urgent. If our expected results find that homes behind destroyed mangroves lost a significant portion of their value relative to others, it will provide compelling evidence that intact ecosystems confer concrete financial benefits to society. Such evidence can catalyze a shift in how decision-makers view conservation as an investment in risk reduction and community resilience. For example, our result in local market-based would confirm the finding Narayan et al. (2019) and Menéndez et al. (2020) that mangroves saved $1.5 billion in property damages during Hurricane and globally provide over $65 billion in flood protection each year. The findings demonstrate that housing markets capitalize these benefits would highlight the hidden costs of habitat degradation and support stronger wetland protection, natural-capital accounting, and investments in coastal resilience.

We acknowledge certain limitations in this design. Our analysis focuses specifically on properties in close distance to mangrove forests where ecological impacts were most visible rather than attempting to fully map how protective benefits decay over long distances inland. Furthermore, while we employ rigorous fixed effects and matching strategies, we remain aware that treated and control areas may differ in unobserved ways that we try address through extensive pre-trend validation. Distinguishing between price drops caused by immediate physical damage versus those caused by updated long-term risk perceptions also presents a challenge. We mitigate this by controlling for precise storm-intensity metrics, though residual unobservables may remain. As the study examines a single storm in a specific ecological and market context, the magnitude of the estimated effects may not generalize to regions with different geographic features, market conditions, or restoration histories. However, the combination of high-resolution ecological data with a robust quasi-experimental strategy allows us to credibly measure the local capitalization of natural infrastructure despite these constraints.

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